

Week 36

Recap on Substitution models, Tree construction methods, and
Molecular clock & MEGA introduction

Questions from you

“More info on distance formulas and how they lead to distance based trees?”

“ Neighbour Joining and UPGMA”

“ How do you choose who is neighbour”

“Topics related to the virus thing” -???? :)

“General Structure of the course, expectations”

Mammal of the Day: Hammer-headed bat

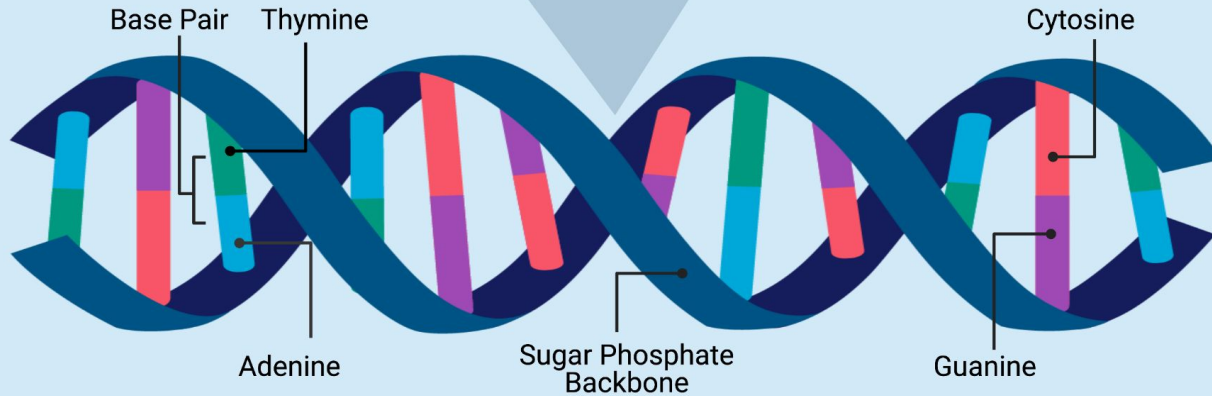
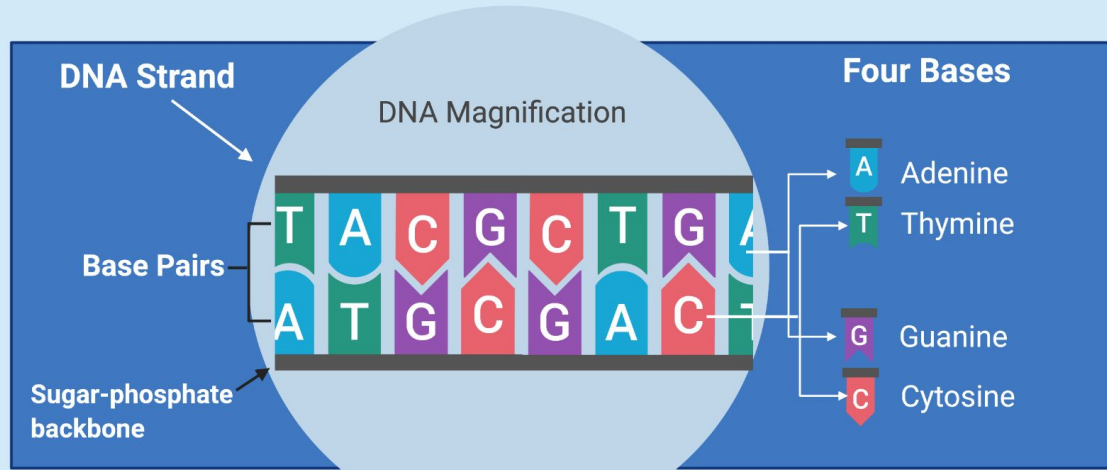
Hypsignathus monstrosus

Central and West Africa

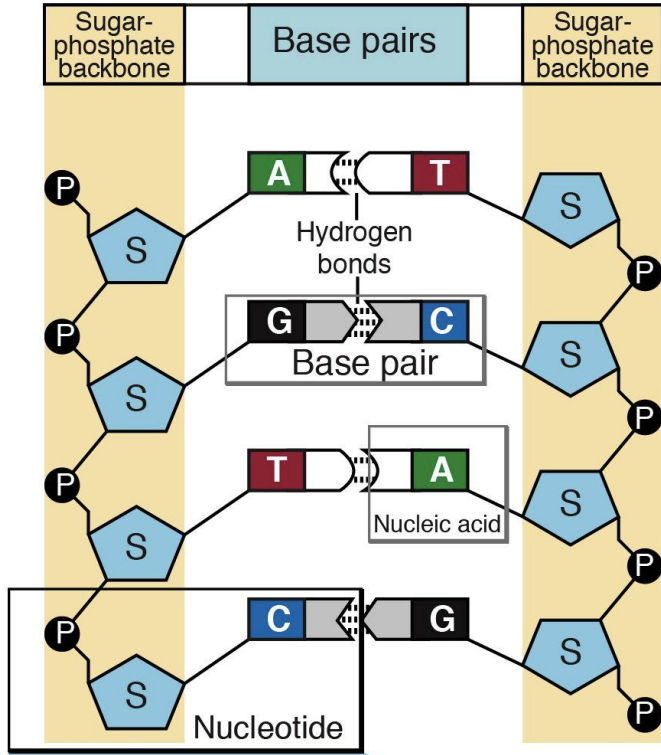
Males have an **enlarged larynx and resonating structures** associated with producing their loud calls.



Molecular Biology Essentials

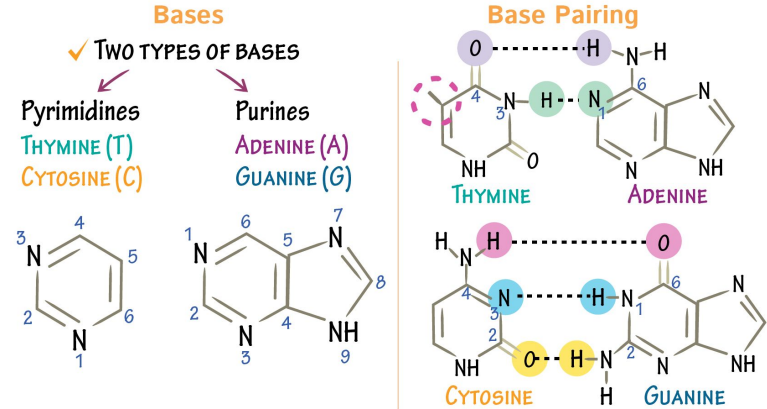


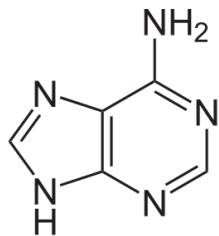
Deoxyribonucleic Acid (DNA)



A Adenine
T Thymine
C Cytosine
G Guanine

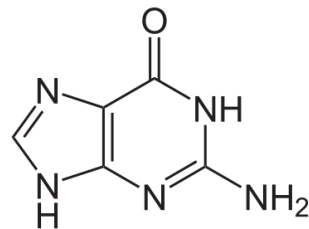
DNA Base Pairing





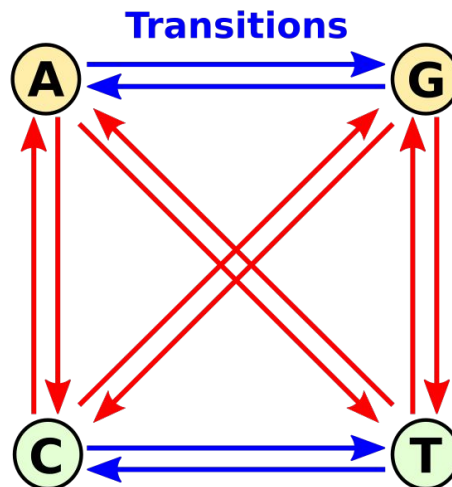
Adenine

Purines



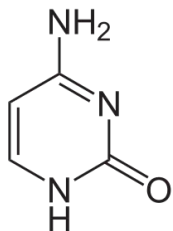
Guanine

Transversions



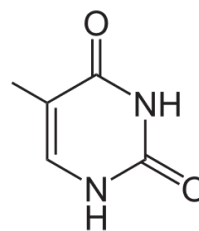
Transversions

Cytosine



Pyrimidines

Thymine



From DNA sequences to Evolutionary Trees

1 Alignment

Align the DNA sequences of the taxa.

2 Observed sequence differences (p)

For each pair, calculate the proportion of different sites (p -distance).

3 Substitution model

Choose a substitution model that describes how nucleotides change.

4 Corrected evolutionary distance (d)

Use the model's formula to convert p into evolutionary distance (substitutions per site).

5 Distance-based tree

Use the distance matrix to infer a tree (e.g., UPGMA or Neighbour Joining).

Example: 4 taxa

	1	2	3	4	5	6	7	8	9	10
A	A	T	G	C	C	A	T	G	T	A
B	A	T	G	T	C	A	T	A	T	A
C	A	C	G	C	T	A	T	G	C	A
D	A	C	A	C	T	G	T	G	C	G

■ Same as A

■ Different from A

Sequences are aligned to identify homologous positions across taxa.

Pairwise p -distance (proportion of sites different)

	A	B	C	D
A	–	0.20	0.30	0.60
B		–	0.20	0.50
C			–	0.40
D				–

p = (number of sites that differ) / (total number of aligned sites)

Example model: Jukes–Cantor (JC69)

Assumptions:

- All nucleotides occur at equal frequencies
- All substitutions occur at the same rate

Substitution rate matrix (Q)

	A	C	G	T
A	–	α	α	α
C	α	–	α	α
G	α	α	–	α
T	α	α	α	–

The model defines the probabilities of change between nucleotides.

Corrected distance (d) using JC69

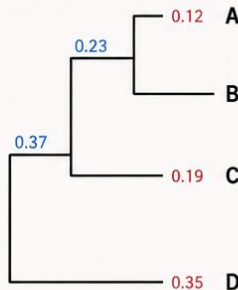
$$d = -\frac{3}{4} \ln \left(1 - \frac{4}{3} p \right)$$

where p = observed proportion of different sites

	A	B	C	D
A	–	0.23	0.37	0.91
B		–	0.23	0.69
C			–	0.56
D				–

Distances are in expected substitutions per site (corrected for multiple hits and model assumptions).

Distance-based tree (example: UPGMA)



The distance matrix is used to infer a tree that best reflects the evolutionary distances among taxa.



Aligned sequences



Count differences
→ p



Choose substitution
model



Calculate corrected
distance d



Build distance-based
tree

1. Substitution models = model of how DNA changes

Their role can vary depending on the phylogenetic method.

For a distance method:

JC69 transforms observed sequence difference **p** into an estimated evolutionary distance **d**

p = observed proportion of aligned sites that are different between two sequences.

Sequence 1	A	T	G	C	G	A	T	T	C	G	A	C	T	G	A
Sequence 2	A	C	G	T	G	G	T	C	C	A	A	C	T	A	A
Match?	✓	✗	✓	✗	✓	✓	✓	✗	✓	✓	✓	✗	✓	✓	✓
	same (match)														
		different (mismatch)													

$$p = \frac{\text{\# of different sites}}{\text{total \# of aligned sites}} = \frac{5}{15} = 0.333$$



Here, $p = 0.333$ means 33.3% of the aligned sites are different between the two sequences.

From observed differences (p) to evolutionary distance (d)

Substitution models correct for multiple substitutions at the same site.

1. Observe aligned sequences

Example alignment (length = 12 sites)

Seq 1	A	T	G	C	G	A	T	T	C	G	A	C	C
Seq 2	A	C	G	T	G	A	C	C	C	G	G	A	T
Match?	✓	✗	✓	✗	✓	✓	✗	✗	✓	✓	✓	✓	✗

☐ same ☒ different

Observed proportion different:

$$p = \frac{\text{\# different sites}}{\text{total sites}} = \frac{5}{12} = 0.417$$

p ignores unseen changes (e.g., A → G → A looks the same as no change).

2. Apply a substitution model

Example: Jukes–Cantor (JC69) model
(equal base frequencies, equal rates)

Relationship between p and distance d
(JC69):

$$d = -\frac{3}{4} \ln \left(1 - \frac{4}{3} p \right)$$

This model estimates the true number of substitutions per site (d).

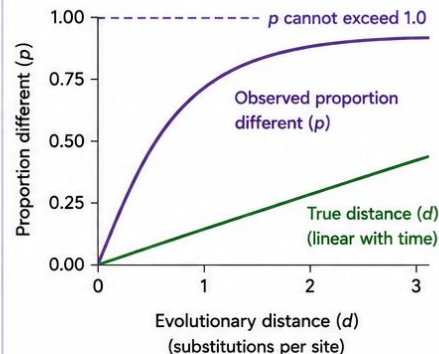
Using our example:

$$p = 0.417$$

$$d = -\frac{3}{4} \ln \left(1 - \frac{4}{3} \times 0.417 \right) = 0.557 \text{ substitutions/site}$$

3. Why p underestimates distance

As sequences diverge, multiple substitutions at the same site become likely.



Substitution models “correct” p for multiple hits and return d .

Summary

Observed data

Aligned sequences
→ count differences

p

Proportion of sites
different

Substitution model

Accounts for multiple
substitutions

d

Evolutionary distance
(substitutions per site)

Use in phylogenetics

Build trees, test clocks,
estimate divergence times



p = what we observe. d = what we estimate after correcting for unseen changes.

Different models (JC69, K80, HKY85, GTR, etc.) make different assumptions and give slightly different d values.

Alignment -> observed sequence differences (p)→substitution model→corrected evolutionary distance

The number of observed differences between two sequences is not always a good measure of how evolutionarily distant those sequences are.

Different nucleotide substitutions may occur with different probabilities, and another is that the same site may have changed more than once during evolution.

What do you think - why?

Substitution models describe our assumptions about how nucleotide changes occur — in other words, how likely it is that one nucleotide is replaced by another over time.

These assumptions are represented by the substitution-rate matrices used by the different models. The chosen substitution model is then used to convert the observed sequence differences into an estimate of **evolutionary distance**, usually expressed as the expected number of substitutions per site.

For example, the **Jukes–Cantor model (JC69)** assumes that all four nucleotides occur at equal frequencies and that every possible nucleotide substitution occurs at the same rate. Under these assumptions, the observed proportion of different sites, p , is corrected using

$$d = -\frac{3}{4} \ln(1 - \frac{4}{3}p)$$

where p is the proportion of aligned sites that differ between the two sequences, and d is the estimated evolutionary distance in substitutions per site.

A: A T G **C** A T **G** T A

B: A T G **T** C A T **A** T A

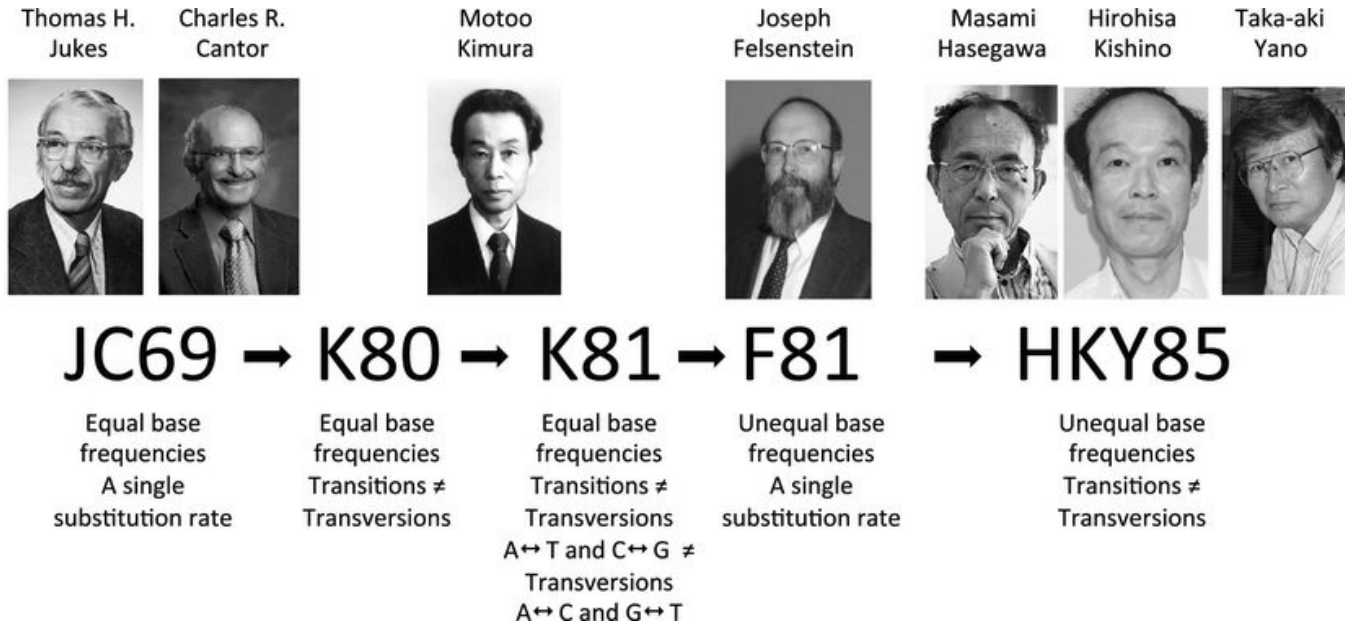
↑ ↑

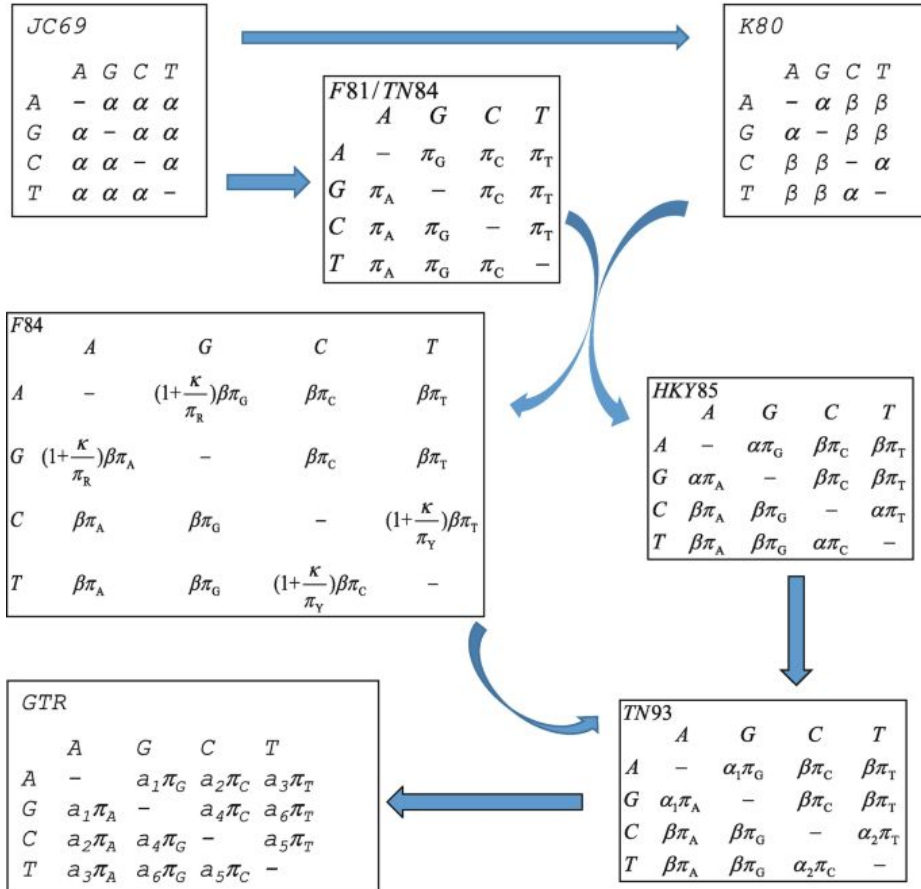
$p = 2$

Different substitution models make different assumptions.

For example, JC69 treats all substitutions equally, whereas the Kimura 2-parameter model allows transitions and transversions to occur at different rates.

Then, once a distance has been calculated for **every pair of sequences**, these distances are collected into a **distance matrix**, which can be used by a tree-construction method such as UPGMA or neighbour joining.





UPGMA and Neighbour Joining in more detail

UPGMA: Step-by-Step Tree Construction

Example with 4 species (A, B, C, D)

1 Start: each species is its own cluster

Clusters: {A}, {B}, {C}, {D}

Distance matrix

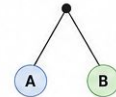
	A	B	C	D
A	–	2	6	6
B	2	–	5	6
C	6	5	–	3
D	6	6	3	–

The matrix contains pairwise evolutionary distances between all clusters (here, individual species).

2 Find the smallest distance

The smallest distance is 2 between A and B.

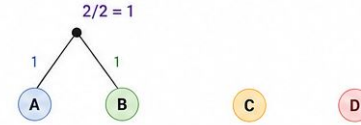
	A	B	C	D
A	–	2	6	6
B	2	–	5	6
C	6	5	–	3
D	6	6	3	–



Join A and B.

3 Create a node by joining closest clusters

Create a new cluster (AB) and place the node halfway between A and B.



Branch lengths from the new node to A and B are each half of the distance between them: $2/2 = 1$.

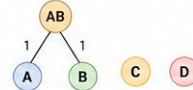
4 Update the distance matrix

Compute distances from the new cluster (AB) to the other clusters using the arithmetic mean.

$$d(AB, C) = (d(A, C) + d(B, C)) / 2 = (6 + 5) / 2 = 5.5$$

$$d(AB, D) = (d(A, D) + d(B, D)) / 2 = (6 + 6) / 2 = 6$$

	AB	C	D
AB	–	5.5	6
C	5.5	–	3
D	6	3	–

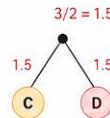


Cluster AB replaces A and B in the matrix. Matrix size is reduced.

5 Find the smallest distance again

The smallest distance is 3 between C and D.

	AB	C	D
AB	–	5.5	6
C	5.5	–	3
D	6	3	–

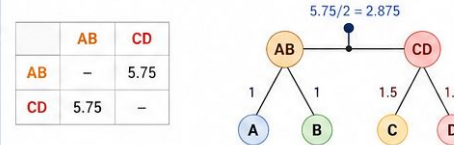


Join C and D. Branch lengths = $3/2 = 1.5$.

6 Update matrix and repeat until one cluster remains

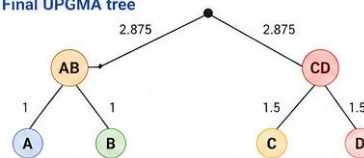
Compute distance between AB and CD:

$$d(AB, CD) = (d(AB, C) + d(AB, D)) / 2 = (5.5 + 6) / 2 = 5.75$$



Join AB and CD. Branch lengths = $5.75/2 = 2.875$.
Now only one cluster remains → the UPGMA tree is complete.

Final UPGMA tree



Key points

- UPGMA assumes a molecular clock (ultrametric tree).
- At each step: find closest clusters → join → update matrix using arithmetic mean → repeat.
- Branch lengths are set so that the distance between joined clusters equals the sum of branch lengths connecting them.

Summary of steps

- 1 Start with all species as individual clusters and compute distance matrix.
- 2 Find the smallest distance in the matrix.
- 3 Join the two closest clusters and create a new node.
- 4 Update the distance matrix using arithmetic means.
- 5 Repeat steps 2–4.
- 6 When only one cluster remains, the tree is complete.

NEIGHBOUR JOINING: STEP-BY-STEP CONSTRUCTION

Goal: build an unrooted tree from a distance matrix by repeatedly joining the pair of taxa that are **close to each other** and, at the same time, **far from all other taxa**.

1 Start with the distance matrix

Each taxon is its own cluster.

Distance matrix (D)

	A	B	C	D	E
A	—	5	9	9	8
B	5	—	10	10	9
C	9	10	—	8	7
D	9	10	8	—	3
E	8	9	7	3	—

Here we have 5 taxa ($n = 5$).

2 Compute N-matrix (neighbour-joining criterion)

1) For each taxon i , compute row sum: $r_i = \sum_k D_{ik}$

$$r_A = 31 \quad r_B = 34 \quad r_C = 34 \quad r_D = 30 \quad r_E = 27$$

2) Compute $N_{ij} = (n-2)D_{ij} - r_i - r_j$ (here $n = 5$, so $n-2 = 3$)
 N_{ij} is large when i and j are close to each other (small D_{ij}) and far from everyone else (large r_i and r_j).

N-matrix

	A	B	C	D	E
A	—	-46	-52	-48	-42
B	-46	—	-56	-54	-49
C	-52	-56	—	-46	-42
D	-48	-54	-46	—	-45
E	-42	-49	-42	-45	—

The maximum N_{ij} is -56 for pair (B, C).

3 Join the nearest neighbours (B and C)

B and C are joined to a new internal node u .

Compute branch lengths from u to B and C:

$$L_{Bu} = \frac{1}{2} D_{BC} + \frac{1}{2(n-2)} (r_B - r_C)$$

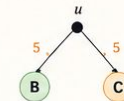
$$L_{Cu} = \frac{1}{2} D_{BC} + \frac{1}{2(n-2)} (r_C - r_B)$$

Here: $D_{BC} = 10$, $r_B = 34$, $r_C = 34$, $n = 5$

$$L_{Bu} = 5 + \frac{1}{6} (0) = 5$$

$$L_{Cu} = 5 + \frac{1}{6} (0) = 5$$

B and C are close to each other (D_{BC} is small) and far from others (large r_B and r_C), so they are joined first.



4 Update the distance matrix

Create a new cluster $u = (B, C)$.

Distance from u to any other taxon k :

$$D_{uk} = \frac{D_{Bk} + D_{Ck} - D_{BC}}{2}$$

Compute:

$$D_{uA} = (5 + 9 - 10)/2 = 2$$

$$D_{uD} = (10 + 8 - 10)/2 = 4$$

$$D_{uE} = (9 + 7 - 10)/2 = 3$$



Now we have 4 clusters: {A}, {D}, {E}, and {u}.

New distance matrix

	A	D	E	u
A	—	9	8	2
D	9	—	3	4
E	8	3	—	3
u	2	4	3	—

5 Repeat: compute N-matrix and join next pair

Row sums:

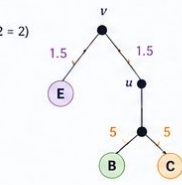
$$r_A = 19 \quad r_D = 16 \quad r_E = 14 \quad r_u = 9$$

$N_{ij} = (n-2)D_{ij} - r_i - r_j$ (now $n = 4$, so $n-2 = 2$)

N-matrix

	A	D	E	u
A	—	-18	-14	-8
D	-18	—	-10	-6
E	-14	-10	—	-4
u	-8	-6	-4	—

Maximum N_{ij} is -4 for pair (E, u).
Join E and u.



Branch lengths:
 $L_{Ev} = 1.5$, $L_{uv} = 1.5$

6 Final update and finish

Update distances (only A and D remain outside):

$D_{AD} = 9$ (unchanged)

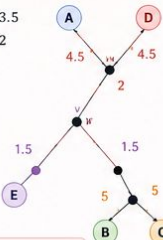
$$D_{Av} = (D_{AE} + D_{Au} - D_{Eu})/2 = (8 + 2 - 3)/2 = 3.5$$

$$D_{Dv} = (D_{DE} + D_{Du} - D_{Eu})/2 = (3 + 4 - 3)/2 = 2$$

Final matrix

	A	D	v
A	—	9	3.5
D	9	—	2
v	3.5	2	—

Join A and D (branch lengths = $9/2 = 4.5$ each).
We now have a single tree. This is the neighbour-joining tree.



Summary of the Neighbour-Joining Algorithm

1 Start with all taxa as single clusters; compute distance matrix D .

2 Compute row sums r_i and the N-matrix: $N_{ij} = (n-2)D_{ij} - r_i - r_j$.

3 Find the pair with the maximum N_{ij} ; they are neighbours.

4 Join them to a new node; compute branch lengths.

5 Update the distance matrix using

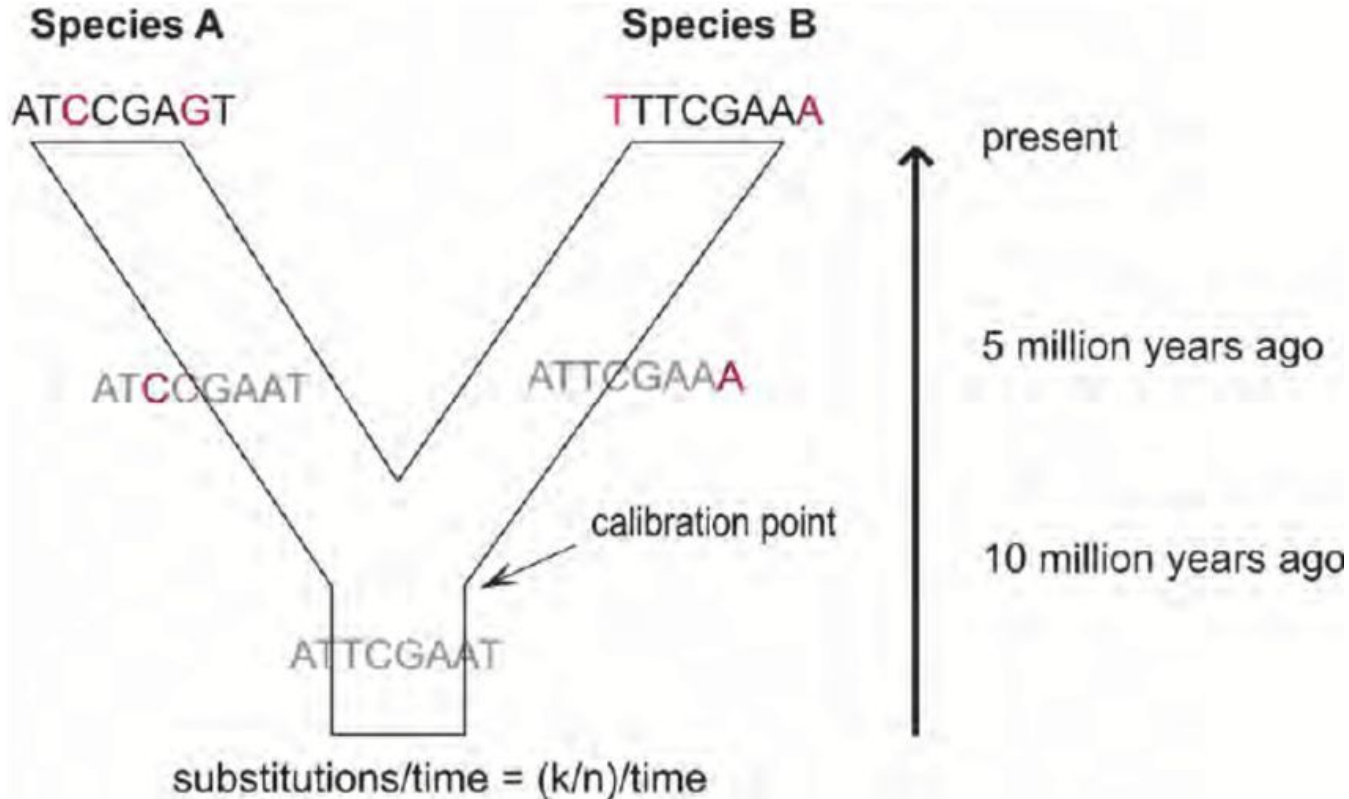
$$D_{uk} = (D_{ik} + D_{jk} - D_{ij})/2.$$

6 Repeat until only two clusters remain; join them.

Key idea

NJ balances being close to each other (small D_{ij}) with being far from all others (large r_i and r_j).

Molecular Clock



Molecular clock = hypothesis that substitutions accumulate at an approximately constant rate over time

Evolutionary distance: D = substitutions per site

For one lineage: $D = rT$

For two lineages that split T years ago:
 $D = 2rT$

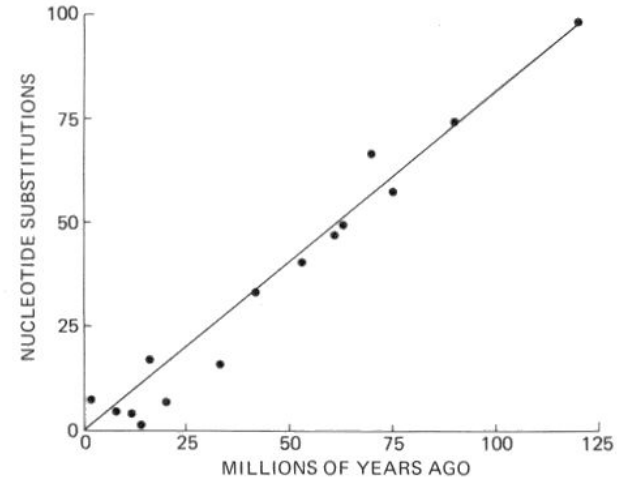
Therefore:
 $T = D/2r$

If divergence time is known from fossils/geology:
 $r = D/2T$

Calibration idea:

known T + observed $D \rightarrow$ estimate $r \rightarrow$ use r to estimate unknown divergence times

Empirical basis: molecular differences often increase roughly with divergence time across taxa.



Does the clock actually hold?

Strict molecular clock predicts equal substitution rates across lineages

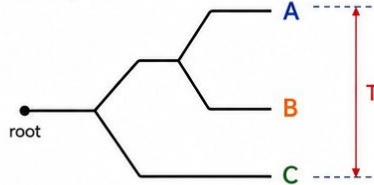
- For sister taxa A and B with outgroup C: $KAC=KBC$, if A and B evolve at the same rate
- If: $KAC \neq KBC$
→ rates differ → strict clock is violated
- **Relative-rate test:** use an outgroup to compare how many substitutions accumulated in two sister lineages
- Primate example in the slides:
 - Human vs New World monkey: $K_{13}=12.98$
 - Old World monkey vs New World monkey: $K_{23}=13.95$
 - Old World monkey lineage evolves faster; reported rate ratio = **1.33**
 - Difference is statistically significant.
- Clock can also be tested with:
 - **Tajima's test**
 - **Likelihood-ratio test** comparing clock vs no-clock models.
- **Connection to UPGMA:** strict clock → ultrametric tree → UPGMA assumption.

Testing the Molecular Clock Hypothesis

Idea: If evolutionary rates are the same, two sister lineages have had the same time to accumulate substitutions.

1. Phylogenetic setup (A, B, C)

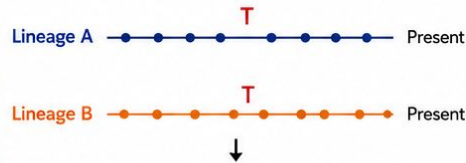
- A and B are sister taxa.
- C is an outgroup.



A and B diverged from their common ancestor the same amount of time ago (T).

2. Prediction under a strict molecular clock

If all lineages evolve at the same rate (r), then A and B have had the same amount of time (T) to accumulate substitutions.

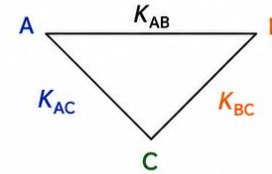


Therefore they should be equally different from the outgroup C.

$$K_{AC} = K_{BC}$$

3. Measuring genetic distances

Estimate the number of substitutions per site between each pair of taxa.



K_{AC} = substitutions per site between A and C

K_{BC} = substitutions per site between B and C

K_{AB} = substitutions per site between A and B

4. Interpreting the result (relative-rate test)

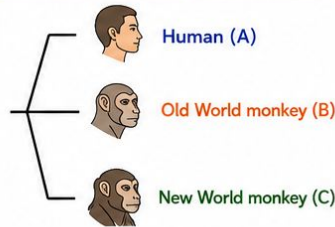
- If $K_{AC} = K_{BC} \rightarrow$ equal rates
Molecular clock holds for A and B.
- If $K_{AC} \neq K_{BC} \rightarrow$ unequal rates
The lineage with the larger distance to C has evolved faster.

Example outcome

Taxon	Distance to outgroup C (substitutions per 100 sites)
A vs C (K_{AC})	12.98
B vs C (K_{BC})	13.95

$K_{BC} > K_{AC} \rightarrow$ Lineage B has accumulated more substitutions $\rightarrow$ B evolves faster.

5. Example from primates (introns)



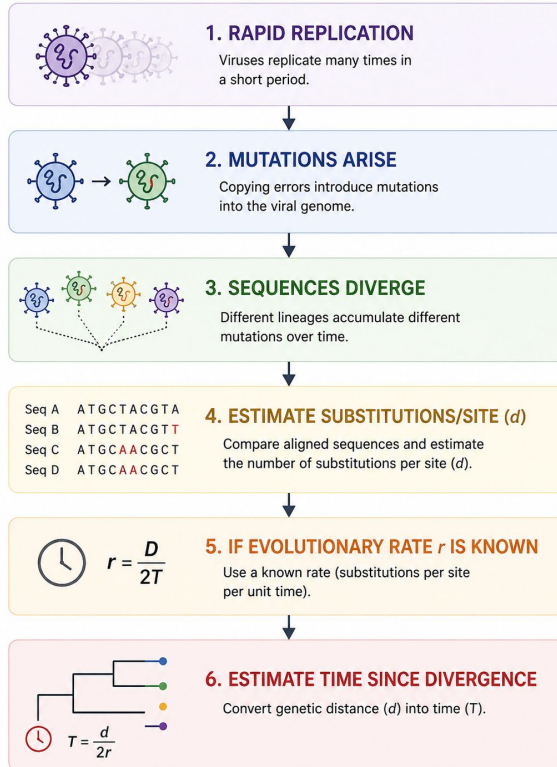
- 29 introns, total length = 15,304 bp
- Distances (substitutions per 100 sites)
 K_{13} (Human–NWM) = 12.98
 K_{23} (OWM–NWM) = 13.95
- Difference: $K_{13} - K_{23} = -0.98$ (significant)
- Rate ratio (B:A) = 1.33

Conclusion: Old World monkey lineage (B) evolves faster than the human lineage (A).
 $\rightarrow$ Strict molecular clock is rejected for these lineages/genes.



Take-home message: Use an outgroup (C) to test whether two sister lineages (A and B) have evolved at the same rate. Equal distances to the outgroup support the molecular clock; unequal distances refute it.

“The virus thing”



Introduction to MEGA12

MEGA is basically a toolbox for going from biological sequences → evolutionary analysis.

For this course, think of it as a graphical interface that lets us perform many of the steps we have just discussed theoretically:

You will need to select:

“Substitution Model/Method”

What assumptions should we make about how sequences evolve?

“Phylogeny / Tree method”

How should we use the sequence information/distances to construct the tree?

MEGA - help: This will teach you everything you need

https://megasoftware.net/web_help_12/index.htm#t=Part_I_Getting_Started%2FA_Walk_Through_MEGA%2FAligning_Sequences.htm

Tips: Mega has its own GTP and multiple video tutorials available on their website:

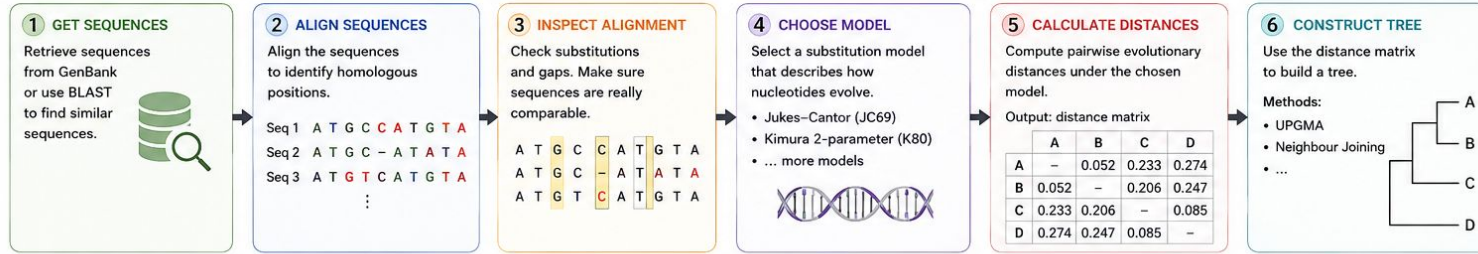
I uploaded a pdf with screenshots included, some of you may find it helpful:)

MEGA: From Sequences to Evolutionary Trees

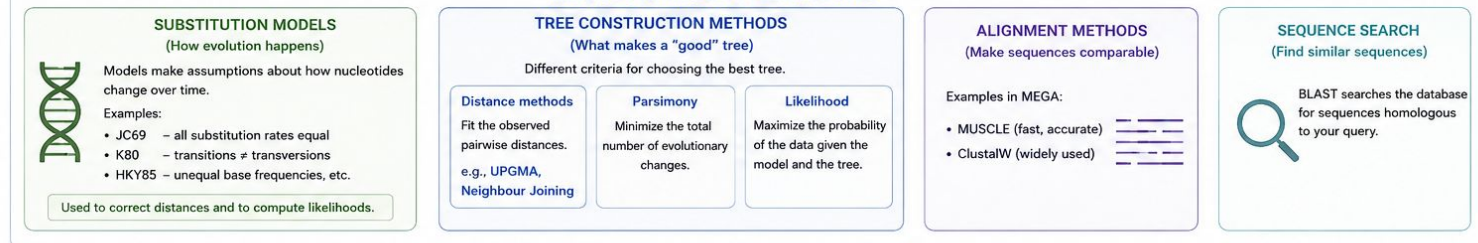
MEGA is a software toolbox for molecular evolutionary analysis.

It implements the methods we learned in class so you can go from biological sequences to evolutionary trees.

THE OVERALL WORKFLOW



METHODS IN MEGA: DIFFERENT ROLES



YOU DECIDE

- Which biological question am I asking?
- Which sequences should I compare?
- Are the sequences homologous?
- Does the alignment look reasonable?
- Which substitution model is appropriate?
- Which tree-building method should I use?
- What assumptions does my method make?
- Does the resulting tree make biological sense?

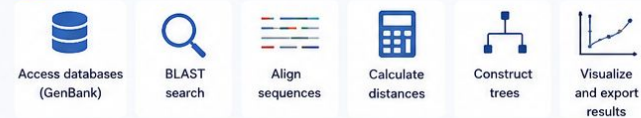
Golden Rule:

Compare like with like.
Make sure the sequences are equivalent (e.g. CDS vs CDS, mRNA vs mRNA).



An output is not the same as a meaningful result.

MEGA DOES



MEGA performs the calculations – you guide the analysis and interpret the results!

THE PIPELINE IN ONE LINE



Today's task:

What are we going to do?

Find biological sequences : Use **GenBank** to retrieve DNA, RNA or protein sequences from different species.

Find related sequences with BLAST: Start with one sequence and search for **similar/homologous sequences** in other organisms.

3. Align sequences in MEGA11: Use **MUSCLE / ClustalW** to identify corresponding positions and inspect:
substitutions • gaps • similarities

4. Estimate evolutionary distances: Calculate pairwise distances between sequences:

aligned sequences→pairwise differences→evolutionary distance matrix